

COLLAB · COLLABORATIVE DEVELOPMENT GROUP

Towards Online Anomaly Monitoring in ROS 2

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The day the robot stopped listening

THE EPISODE

01 **It lagged. It stopped answering It
nearly hit a shop window**

02 **Nothing had crashed**
Every node alive, every topic there

03 **We found it in the logs**
Many days later



A node can fall ill in silence

THE PROBLEM



Tens of independent nodes on one graph. **One is getting worse, and nothing crashes**

WHAT YOU READ

**A snapshot,
not the story**

WHAT YOU LACK

**No one knows which
node is guilty**

WHAT YOU CANNOT DO

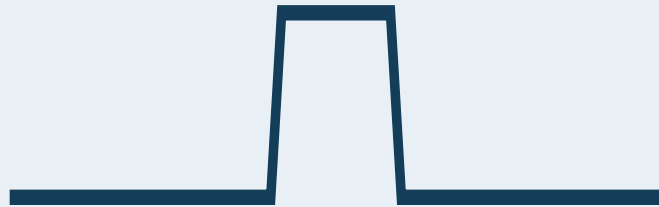
Fix the problem



The anomalies

THE PROBLEM

Step



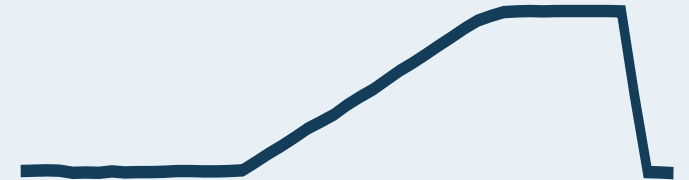
One quantity jumps, all at once

Correlation



Two quantities stop moving together, and both still look normal alone

Drift



It drifts for minutes, and never jumps

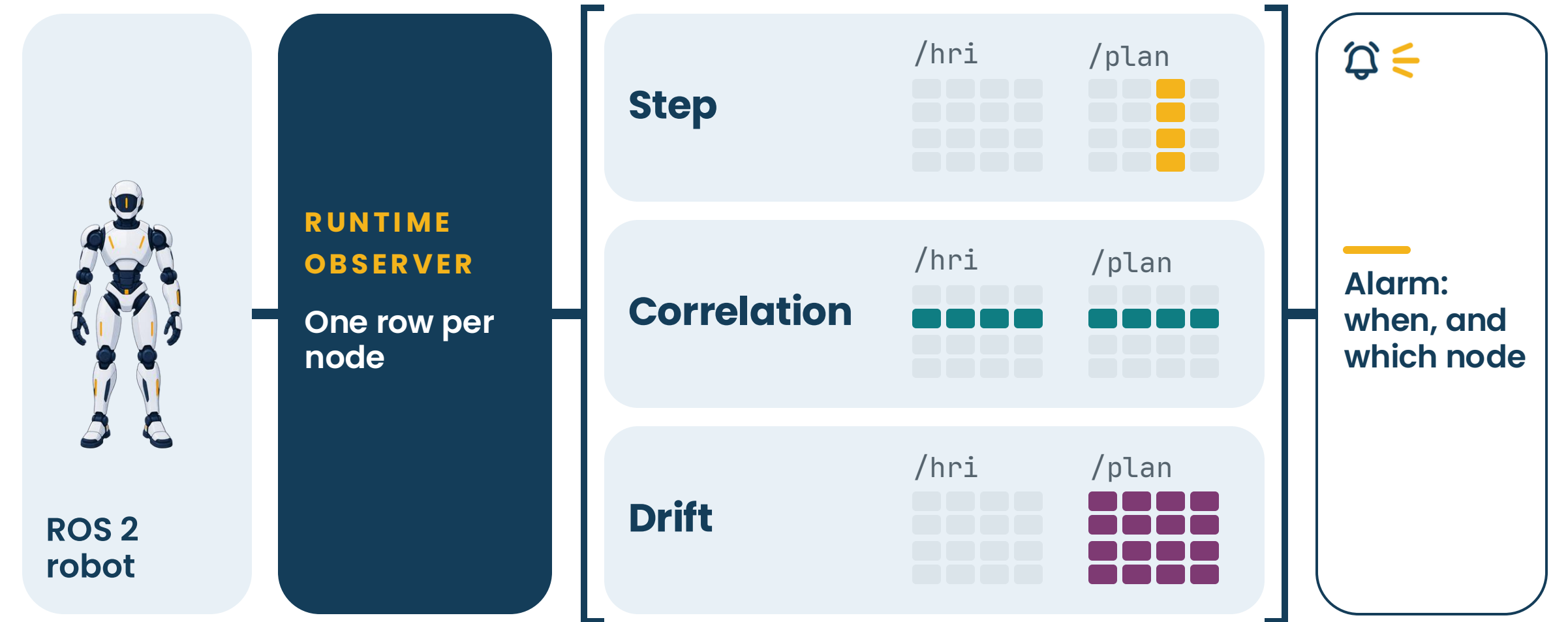


One runtime observer, three ways of looking

THE APPROACH

One runtime observer, three ways of looking

THE APPROACH



The observer: two ways to measure a node

THE DATA

— INSIDE ROS 2 · TRACING

What the software did

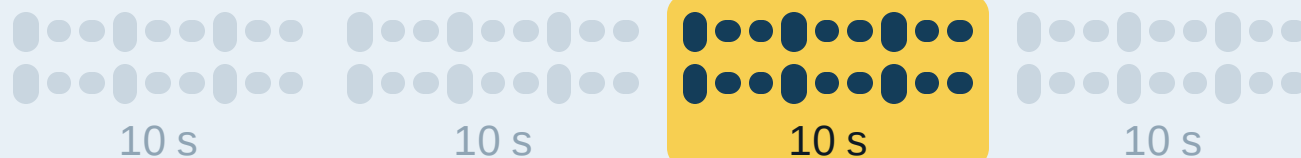
- every callback
- every message published



— OUTSIDE · THE SYSTEM

What the machine paid

- CPU, memory, threads
- context switches, disk



aggregated
→

ONE ROW PER NODE



What we inject

THE DATA

Steps

18 6 sim/robot boots
20 to 90 s

CPU starvation, freezes

Broken correlations

16 6 sim/robot boots
300 s each

latency, CPU/msg,
publish ratio, threads, links

Slow drift

10 6 sim/robot boots
20 min each

CPU quota tightened +3% a
minute

CLEAN GROUND **416** clean windows in 6 different boots, nothing injected

105 h
of recording

1.2 M
node-rows

80 / 25
hours, sim / robot

22 / 78
node, sim/ robot



Step: four anomaly detectors

WHERE WE STAND

— robust band

median and robust deviation

— z-score

three standard deviations

— SPOT

the extreme-value tail

— ADWIN

the whole distribution moved

WHERE A FAULT WAS PRESENT

15 of 18 faults detected

zero alarms before onset, in all 18

FALSE POSITIVES

12 of 416

clean windows flagged, where nothing was injected



Correlation: how well it sorts

WHERE WE STAND

TOP 5 OF 12

pca_node_fusion

maha_node

pca_global

dense_ae_global

dense_ae_pooled

HOW WELL IT SORTS *



0.88



0.86



0.79



0.76



0.72

FALSE ALARMS

one every 3.1 min

one every 5.6 min

one every 4.1 min

one every 4.7 min

one every 6.0 min

ecod_control

one metric at a time



0.61

one every 5.2 min

* Mean VUS-PR over 6 boots, where 0 is bad and 1 is perfect · the blind control reads no correlation



Drift: 10 faults detected of 10 injected

WHERE WE STAND

TOP 5 OF 20

one node, 30 windows of history, sliding by 5

<code>iforest_mean_dev</code>	● ● ● ● ● ● ● ● ● ●	10 of 10
<code>lstm_ae_forecast</code>	● ● ● ● ● ● ● ● ● ●	10 of 10
<code>pca_reconstruction</code>	● ● ● ● ● ● ● ● ● ○	9 of 10
<code>maha_window</code>	● ● ● ● ● ● ● ● ● ○	9 of 10
<code>lstm_ae_reconstruction</code>	● ● ● ● ● ● ● ● ● ○	9 of 10

never the wrong node

no alarm on clean runs



What comes next

FUTURE WORK

NEXT **The full campaign on the physical robot**

THEN **Fine-tune Correlation and Drift**

AFTER **Beyond one robot** TIAGo Pro, TIAGo, R1, Go2: two vendors, three body plans



What comes next

FUTURE WORK

NEXT

The full campaign under the randomised protocol

On the physical robot, with 6 repetitions instead of 4.

THEN

Fine-tune Correlation and Drift

Every model here is fitted on normal data only. On a bench where faults fall at a predictable time, a model allowed to see them learns *when*, not *what*.

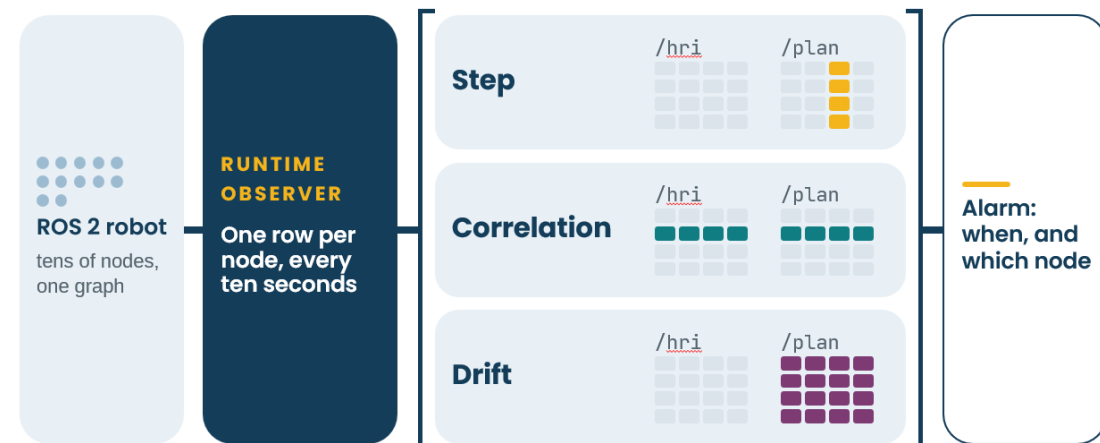
AFTER

Beyond one robot, and what it costs

TIAGo Pro, TIAGo, R1, Go2: two vendors, three body plans

One runtime observer, three ways of looking

THE APPROACH



We inject faults that cause failures

THE ANOMALIES

Three kinds of anomaly, and each one fails in a different way

